

INTRODUCTION TO CYBERSECURITY

📌 MODULE 1 — INTRODUCTION TO CYBERSECURITY

Cisco Networking Academy

Cisco Skills for All

Course Screenshots → Digital

COURSE

Introduction to
Cybersecurity

MODULE

1 of 5

STARTED

17 / 08 / 2026

FORMAT

Notes + Quiz + Test

0 Module Introduction

- **5 modules + final exam**, ~6 hours total
- Self-paced — progress auto-saves; resume anytime on desktop, laptop, tablet, or mobile
- Includes videos, interactive activities, knowledge checks, and quizzes

On completion, learners can:

- Explain the basics of being safe online, and its potential impact
- Explain the most common cyber threats, attacks, and vulnerabilities
- Explain how organizations protect their operations against these attacks
- Access resources to explore career options in cybersecurity

1 1.1 The World of Cybersecurity

1.1.1 What Is Cybersecurity?

- **Cybersecurity** = ongoing effort to protect individuals, organizations, and governments from digital attacks
- Works by protecting networked systems and data from unauthorized use or harm
- Protection operates across multiple levels — personal, organizational, government

1.1.2 Protecting Your Personal Data

- Personal data = any info that can identify a person; exists both **offline** and **online**
- Real-world traits form the **offline identity**
- Username and online behavior form the **online identity**

1.1.3 Your Online Identity

- Exercise: choosing a safe username for a new online identity

- **Safe:** "j.doe12", "jdoe" — no full real name or birth year
- **Unsafe:** "jane.doe", "jdoe1990" — reveals identifying info

1.1.4 Your Data

- Personal data includes: name, social security number, driver's license number, date/place of birth, mother's maiden name, photos/messages
- Cybercriminals can use this to identify and impersonate a person
- Result: infringed privacy, damaged reputation

1.1.5 Where Is Your Data?

- Scenario: a work photo shared with 5 close friends worldwide



Once shared, data can spread far beyond the original owner's control.

1.1.6 What's More...

- **Medical records:** doctor updates record → shared with insurance company for billing
- **Store loyalty cards:** save money, but build a purchasing-behavior profile used for targeted marketing

1.1.7 Smart Devices

- Devices no longer just access personal data (bank statements, mobile banking) — they now **generate** data too
- Wearables (smartwatches, activity trackers) collect data for clinical research, health monitoring, fitness tracking
- Growing fitness-tracker market = growing risk to personal data

1.1.8 What Do Hackers Want?

- Primarily: money
- Reflective exercise: recall real examples of financial cybercrime

1.1.9 Identity Theft

- **Medical theft:** stolen medical insurance used for fraudulent procedures — recorded in the victim's own medical record
- **Banking:** stolen credentials → access bank/credit accounts → file fake tax returns, take loans in victim's name → ruined credit rating

1.1.10 Who Else Wants My Data?

Entity	Why they want your data
🌐 Internet Service Provider (ISP)	Tracks online activity; may sell data to advertisers; may be legally required to share info with government surveillance agencies/authorities
👤 Advertisers	Monitor shopping habits & preferences; send targeted ads
🔍 Search engines & social media	Gather gender, geolocation, phone number, political/religious ideology from search history; sold to advertisers
🌐 Websites you visit	Use cookies to track activity for personalization; data trail often ends up with advertisers

2 1.2 Organizational Data

1.2.1 Types of Organizational Data

Traditional data:

- **Transactional:** buying/selling, production, employment decisions
- **Intellectual property:** patents, trademarks, product plans — trade secret; disastrous if lost
- **Financial:** income statements, balance sheets, cash flow statements

IoT and Big Data:

- IoT = large network of connected physical objects (sensors, software) that collect/share data
- Cloud storage growth + IoT growth = exponential data increase → "Big Data"

1.2.2 The Cube (The McCumber Cube)

- Created by John McCumber, 1991
- Framework to evaluate information security initiatives across 3 dimensions

Dimension 1 — Foundational principles (CIA Triad)

Principle	Meaning
🔒 Confidentiality	Prevents unauthorized disclosure of sensitive info. Methods: data encryption, identity proofing, two-factor authentication
✓ Integrity	Protects data/processes from intentional/accidental modification. Methods: hash functions, checksums
🌐 Availability	Authorized users can access systems/data when needed. Achieved via equipment maintenance, hardware repairs, updates, backups

Dimension 2 — States of data

State	Meaning
⚙️ Processing	Data being used/operated on (data in process)
💾 Storage	Data at rest in memory or storage devices
📡 Transmission	Data in transit between systems

Dimension 3 — Security measures

Measure	Meaning
🧠 Awareness, training & education	Users know potential threats and protective actions
⚙️ Technology	Software/hardware solutions (e.g., firewalls) protecting systems
📋 Policy & procedure	Administrative controls — incident response plans, best-practice guidelines

1.2.3 What Do You Think? (Phishing Exercise)

- Scenario: suspicious "@pollo" email — claims restricted access, asks to click a link to "reactivate"

Signs of phishing: the link URL, language/spelling/grammar, graphics, and the sender's email address

1.2.4 Is This for Real? (Razer Case Study)

- August 2020: Razer breach exposed ~100,000 customers' personal data
- Cause: misconfigured cloud cluster exposed to public internet

- Took 3+ weeks to secure; exposed data usable for social engineering/fraud
- Lesson: proactive cloud security is essential

1.2.5 Data Security Breaches

- **Persirai botnet (2017):** targeted 1,000+ IP camera models via open ports; malware self-deleted to avoid detection; 122,000+ cameras hijacked for DDoS attacks without owners' knowledge
- **Equifax (Sept 2017):** web app vulnerability exposed millions' data; response site used a new domain (not an equifax.com subdomain), letting criminals build lookalike phishing sites

1.2.6 Consequences of a Security Breach

Consequence	Impact
📉 Reputational damage	Long-term trust loss; customers may leave, employees may resign
✍️ Vandalism	Hacker alters website content/contact info
👤 Theft	Stolen data exploited for money/identity
🕒 Loss of revenue	Downtime, lost business, security investment, fines
📁 Damaged intellectual property	Loss of trade secrets/confidential docs hurts competitiveness

3 1.3 What Was Taken?

1.3.1 - 1.3.2 Scenario 1: Hotel Data Breach (*fictional*)

- Global hotel chain: 3M+ guests' data exposed
- Cause: hackers used a stolen employee login
- Passwords/financial info not believed compromised

What was taken: names, email addresses, phone numbers of 3M+ guests

1.3.3 - 1.3.4 Scenario 2: eLearning Platform at Risk

- eLearning platforms = growing target as digital learning expands

Exploit used: weaknesses in the platform's security practices

1.3.6 Key Takeaways

- Security breach = unauthorized access to data/apps/services/devices, exposing info attackers exploit for gain
- Staying vigilant and aware of common threats helps avoid becoming a victim

4 1.4 Cyber Attackers

1.4.1 Types of Attackers

- Attackers exploit vulnerabilities for personal/financial gain — after "everything" from credit cards to product designs
- Categorized as: **white hat, gray hat, black hat**

1.4.2 What Color Is My Hat? (Scenarios)

Scenario	Hat
Hacked ATM systems remotely, then worked with manufacturers to fix the vulnerabilities	👤 Gray hat
Transferred \$10M using stolen customer PIN/account credentials	👤 Black hat
Job is to identify weaknesses in a company's system	👤 White hat
Used malware to steal credit card data, sold to highest bidder	👤 Black hat
Found a vulnerability while researching a network they were authorized to access	👤 White hat

1.4.3 Internal and External Threats

Type	Description
🏢 Internal	• Mishandle confidential data • Facilitate attacks via infected USB media • Invite malware via malicious emails/websites • Threaten internal servers/network infrastructure
🌐 External	• Exploit network vulnerabilities • Gain unauthorized device access • Use social engineering to access organizational data

1.4.4 What Do You Think?

- Phishing traced to a former employee whose access wasn't fully revoked
- They logged into the customer database after leaving to join a competitor

Classification: Internal threat — retained access makes it internal, despite no longer being an employee

5 1.5 Cyberwarfare

1.5.1 Sign of the Times (Stuxnet)

- **Cyberwarfare** = using technology to penetrate/attack another nation's systems to cause damage or disrupt services (e.g., shutting down a power grid)
- **Stuxnet:** state-sponsored malware designed to cause physical damage to computer-controlled equipment
- Famous case: impact on Iran's nuclear enrichment infrastructure
- *Referenced video: "Stuxnet: Anatomy of a Computer Virus"*

1.5.2 The Purpose of Cyberwarfare

- Main goal: gain advantage over adversaries (nations or competitors)
- Can destabilize a nation, disrupt commerce, undermine citizens' trust in government
- All without the attacker physically entering the target country

6 1.6 Quiz — Knowledge Check

Q1. An individual user profile on a social network site is an example of an ___ identity.

Ans — Online.

Q2. What level of cyber protection does each factor require?

- Your online identity → Personal
- A customer database → Organizational
- Economic stability → Government

Q3. Which pieces of information are personal data? (select three)

Ans — Date and place of birth, Social security number, Driver license number.

Q4. A neighbor with no social media, only browsing — do they have no online identity?

Ans — No. Browsing alone still creates an online identity.

Q5. Which methods ensure confidentiality? (select three)

Ans — Username ID and password, Data encryption, Two-factor authentication.

Q6. Foundational principles of the McCumber Cube? (select three)

Ans — Confidentiality, Availability, Integrity.

Q7. Key motivation of a white hat attacker?

Ans — Discovering weaknesses in networks/systems to improve their security.

Q8. Why can internal threats cause greater damage than external threats?

Ans — Internal users have direct access to infrastructure devices.

Q9. Why is each organization interested in your online identity?

- ISPs: may be legally required to share data with government surveillance agencies/authorities
- Advertisers: gather data from your online activity, share/sell for profit
- Social media platforms: track activity via cookies for personalization
- Websites: monitor activity to send targeted ads

Q10. Which statement correctly describes cybersecurity?

Ans — The ongoing effort to protect individuals, organizations, and governments from digital attacks.

80%

Congratulations, you have passed the exam. ✓ Module 1 Knowledge Check — Complete

INTRODUCTION TO CYBERSECURITY

♥ MODULE 2 — ATTACKS, CONCEPTS AND TECHNIQUES

Cisco Networking Academy

Cisco Skills for All

Course Screenshots → Digital

COURSE

Introduction to
Cybersecurity

MODULE

2 of 5

STARTED

18 / 08 / 2026

FORMAT

Notes + Quiz + Test

0 2.1 Analyzing a Cyber Attack

2.1.1 Types of Malware

- Malware = any code used to steal data, bypass access controls, or harm/compromise a system

Type	What it does
 Spyware	Tracks/spies on activity, logs keystrokes, captures sensitive data (e.g., online banking) by modifying security settings; often bundles with legitimate software or Trojans
 Adware	Automatically delivers ads (usually via browser); often comes bundled with spyware
 Backdoor	Bypasses normal authentication for unauthorized access; allows remote access and remote commands; runs in background, hard to detect
 Ransomware	Holds a system/data captive (usually via encryption) until payment is made; often spread via phishing emails or software vulnerabilities
 Scareware	Uses fake OS-style pop-ups warning of a risk, tricking the user into running a program that then infects the system
 Rootkit	Modifies the OS to create a backdoor for remote access; exploits vulnerabilities for privilege escalation; can modify forensic/monitoring tools, very hard to detect; infected systems often need a full wipe
 Virus	Replicates and attaches itself to executable files; needs user interaction to activate; can be harmless or destructive; can mutate to avoid detection; spreads via USB, optical disks, network shares, email
 Trojan horse	Masks its true intent as legitimate software; exploits user privileges; found in image/audio files or games; does not self-replicate, acts as a decoy
 Worms	Self-replicating, spreads computer-to-computer without a host program or user participation; exploits system vulnerabilities; carries a malicious payload; the 2001 Code Red worm infected 300,000+ servers in 19 hours

2.1.2 Symptoms of Malware

- Increased CPU usage, slowing down the device
- Decrease in web browsing speed
- Unexplainable problems with network connections
- Modified or deleted files
- Presence of unknown files, programs, or desktop icons
- Unknown processes running
- Programs turning off or reconfiguring themselves
- Emails being sent without the user's knowledge or consent

2.1.3 What Do You Think? (Matching Exercise)

Description	Malware
Tracks online activity and captures data	Spyware
Automatically delivers advertisements	Adware
Holds a system captive until payment is made	Ransomware
Attaches to legitimate programs, spreads via USB/optical media/network shares/email	Virus
Replicates independently by exploiting network vulnerabilities	Worms

1 2.2 Methods of Infiltration

2.2.1 Social Engineering

- Manipulating people into performing actions or divulging confidential information
- Relies on people's willingness to help, but preys on weaknesses too

- Example: attacker calls with an "urgent problem," appeals to vanity/greed, or name-drops to invoke authority and gain network access

Type	Description
Pretexting	Attacker calls an individual and lies to them to gain access to privileged data — e.g., pretending to need personal/financial data to "confirm their identity"
Tailgating	Attacker quickly follows an authorized person into a secure, physical location
Something for something (quid pro quo)	Attacker requests personal information from a person in exchange for something, like a free gift

2.2.2 Denial-of-Service (DoS)

- Simple to carry out, even by an unskilled attacker; interrupts network service to users/devices/applications

Type	Description
👤 Overwhelming quantity of traffic	Network/host/application sent an enormous amount of data at a rate it can't handle — slows or crashes the target
📧 Maliciously formatted packets	Packets with errors/improper formatting that the receiver can't handle — causes the device to run slowly or crash

2.2.3 Distributed DoS (DDoS)

- Similar to DoS, but originates from multiple coordinated sources
- Attacker builds a botnet of infected "zombie" hosts, controlled by handler systems
- Zombies keep scanning and infecting more hosts, growing the botnet
- Hacker instructs the handlers to make the zombie botnet carry out a DDoS attack

2.2.4 Botnet

- Bot = infected via unsafe websites, infected email attachments, or infected media files
- Botnet = group of bots (tens/hundreds of thousands), controlled via a command-and-control server
- Used to: distribute malware, launch DDoS attacks, distribute spam, run brute-force password attacks; often rented out to third parties

Cisco's botnet detection process:

- Infected bots try to communicate with a command-and-control host on the internet
- Cisco Firewall botnet filter detects traffic from devices infected with known malicious botnet code
- Cloud-based Cisco Security Intelligence Operations (SIO) pushes updated filters matching new known botnets
- Alerts go to Cisco's internal security team to prevent, mitigate, and remedy infected devices

2.2.5 On-Path Attacks

- Attacker intercepts/modifies communication between two devices, to collect info or impersonate one of them
- Also called man-in-the-middle or man-in-the-mobile attacks

Type	Description
👤 Man-in-the-Middle (MitM)	Cybercriminal takes control of a device without the user's knowledge; intercepts/captures user info before it reaches its destination; often used to steal financial info
📱 Man-in-the-Mobile (MitMo)	Variation of MitM targeting mobile devices; instructs the infected device to exfiltrate sensitive info; e.g., ZeuS malware quietly captures two-step verification SMS codes

2.2.6 SEO Poisoning

- SEO = improving a website's visibility in search engine results
- Attackers exploit popular search terms to push malicious sites higher in rankings — called **SEO poisoning**
- Common goal: increase traffic to malicious sites that host malware or attempt social engineering

2.2.7 Wi-Fi Password Cracking

A colleague asks for the private Wi-Fi password, claiming their public Wi-Fi isn't working.

Correct response: "I'm not sure we're allowed to use the private Wi-Fi network. Let me check with my manager first."

2.2.8 Password Attacks

Attack	Description
🔑 Password spraying	Sprays a few commonly-used passwords (e.g., 'Password123') across many accounts before trying the next common password — avoids frequent lockouts, stays undetected
📖 Dictionary attacks	Systematically tries every word in a dictionary or common-word list as a password
🔪 Brute-force attacks	Tries all possible combinations of letters, numbers, and symbols until the password is found — simplest and most common method
🌈 Rainbow	Uses a rainbow table (large dictionary of precomputed hashes and their source passwords); compares a password's hash to

attacks	the table instead of calculating each hash like brute-force
👁 Traffic interception	Plain-text/unencrypted passwords can be read by intercepting communications; anyone with account/device access can read a password stored in clear text

2.2.9 Cracking Times

- Exercise: order passwords by brute-force cracking time (shortest to longest)
- Order: "Password" → "3trawberry" → "K4km9n2R" → "H\$1gh#7iD@3" (weakest to strongest)

2.2.10 Advanced Persistent Threats (APT)

- Multi-phase, long-term, stealthy, advanced operation against a specific target
- Individual attackers usually lack the skill/resources/persistence to run APTs
- Usually well-funded; targets organizations or nations for business/political reasons
- Main purpose: deploy customized malware on target systems and remain undetected

2.2.11 It's Over to You (Scenario Matching)

Scenario	Attack Type
A stranger asks you to hold the door — they "forgot their access card"	Social engineering
Error message: "Your connection was interrupted. A network change was detected."	DoS
Top Google result for the company site redirects to an antivirus ad page	SEO poisoning

2 2.3 Security Vulnerability and Exploits

- **Security vulnerability:** any software or hardware defect
- **Exploit:** a program written to take advantage of a known vulnerability
- **Attack:** using an exploit against a vulnerability to gain access to a system, its data, or a resource

2.3.1 Hardware Vulnerabilities

- Usually result from hardware design flaws
- **Rowhammer:** RAM capacitors installed close together can influence neighbors when repeatedly accessed ("hammered"), corrupting data in nearby memory
- **Meltdown & Spectre:** discovered by Google researchers; affect almost all CPUs made since 1995 (desktops, laptops, servers, smartphones, smart devices, cloud); allow attackers to read all memory (Meltdown) or data from other apps (Spectre) — known as side-channel attacks
- Hardware vulnerabilities are device-model specific, not randomly exploited; traditional malware protection + good physical security are enough for everyday users

2.3.2 Software Vulnerabilities

- Usually introduced by errors in OS or application code
- **SYNful Knock (2015):** vulnerability in Cisco IOS; let attackers control enterprise-grade routers (e.g., legacy Cisco ISR) to monitor network traffic and infect other devices
- Introduced via an altered IOS image installed on routers — mitigated by verifying IOS image integrity and restricting physical access to authorized personnel only

2.3.3 Categorizing Software Vulnerabilities

Category	Description
📦 Buffer overflow	Data written beyond a buffer's memory limits lets an app access memory allocated to other processes — can cause a crash, data compromise, or privilege escalation
✖ Non-validated input	Malicious input data forces a program to behave unexpectedly — e.g., a crafted image with invalid dimensions forcing incorrect buffer sizes
🕒 Race conditions	Vulnerability arises when required ordered/timed events don't occur in the correct order or at the proper time
🔧 Weaknesses in security practices	Rolling custom security/encryption algorithms instead of using tested, verified libraries tends to introduce new vulnerabilities
🔑 Access control problems	Improper use of access controls (who can do what) creates vulnerabilities; physical access can bypass nearly all controls — encryption + restricted physical access are essential

2.3.4 Software Updates

- Goal: stay current and avoid exploitation of known vulnerabilities
- OS makers (Microsoft, Apple, etc.) release patches/updates regularly; apps like browsers are updated by their own vendors
- Organizations often use third-party security researchers or in-house penetration testing teams to find and patch vulnerabilities before they're exploited
- Example: Google's **Project Zero** — a permanent team dedicated to finding software vulnerabilities

Real-world example — CVE-2026-56914 (Pixel GXP driver, found via Google Project Zero):

- **Bug:** the GXP camera driver function `gcip_iommu_mapping_buffer_flush_sgt()` called `set_page_dirty()` directly on a memory page that was neither locked nor reference-tracked
- `set_page_dirty()` requires the caller to already hold a lock that stops the page from being truncated/freed while it's marked dirty — this function held no such lock
- **Impact:** the page could be freed while still marked dirty — a **use-after-free (UAF)** vulnerability in the kernel, reachable through `ioctl()` calls on `/dev/gxp` (accessible to several SELinux contexts, including the camera app)
- **Fix:** replace `set_page_dirty()` with `set_page_dirty_lock()` — this variant automatically locks the page/folio first, so it cannot be freed out from under the driver while marked dirty
- Classified S2/P2 severity, 90-day disclosure deadline (Project Zero's standard policy) — a real illustration of category "buffer/memory handling" software vulnerabilities and why security patching matters

2.3.5 What Do You Think? (Scenario Matching)

Scenario	Category
Network password emailed in plain text, no prompt to change it	Weakness in security practice
Past employees still have access to the customer database	Access control problem
New users can log in even with an incorrectly formatted email address	Non-validated input

3 2.4 The Cybersecurity Landscape

2.4.1 Cryptocurrency

- Digital money used to buy goods/services, secured using strong encryption techniques
- Banks, governments, and companies (Microsoft, AT&T) are increasingly engaging with it

2.4.2 Cryptojacking

- Emerging threat that hides on a device (computer, phone, tablet, laptop, server)
- Uses the machine's resources to "mine" cryptocurrency without the user's consent or knowledge
- Many victims don't realize they've been hacked until it's too late

4 2.5 Quiz — Knowledge Check

Q1. What is the purpose of a rootkit?

Ans — To gain privileged access to a device while concealing itself.

Q2. Which example illustrates how malware might be concealed?

Ans — An email is sent to employees with an attachment that looks like an antivirus update, but the attachment actually consists of spyware.

Q3. What is the primary goal of a DoS attack?

Ans — To prevent the target server from being able to handle additional requests.

Q4. What is the most common goal of SEO poisoning?

Ans — To increase web traffic to malicious sites.

Q5. A set of changes made to a program/application to update, fix, or improve it is referred to as what?

Ans — A patch.

Q6. Which should be carried out to address known software vulnerabilities of a specific application?

Ans — Install a security patch.

Q7. "Securing physical access to target equipment is an organization's best defense against a cyber attack." True or false?

Ans — True.

Q8. What is a miner?

Ans — A person that solves complex mathematical puzzles to verify a transaction.

INTRODUCTION TO CYBERSECURITY

♥ MODULE 3 — PROTECTING YOUR DATA AND PRIVACY

Cisco Networking Academy

Cisco Skills for All

Course Screenshots → Digital

COURSE

Introduction to
Cybersecurity

MODULE

3 of 5

STARTED

19 / 08 / 2026

FORMAT

Notes + Quiz + Test

0 3.1 Protecting Your Devices and Network

3.1.2 Protecting Your Computing Devices

Tip	Details
Turn the firewall on	Use at least one firewall type (software or a router's hardware firewall); keep it on and updated to block unauthorized access
Install antivirus and antispyware	Malicious software can destroy data, take over a device, or spy on activity; only download from trusted sites and keep antivirus/antispyware updated
Manage OS and browser	Set security settings to medium or higher; regularly update the OS and browser with the latest patches
Set up password protection	Password-protect all devices (PCs, laptops, tablets, phones); encrypt sensitive data; store only necessary info on mobile — a compromised device can expose all cloud-linked data (e.g., iCloud, Google Drive)

IoT device risk: IoT devices rarely get software updates, so known vulnerabilities often stay unpatched. Best practice: isolate IoT devices on a separate network. **Shodan** is a web-based IoT device scanner that helps identify vulnerable devices online.

3.1.3 Wireless Network Security at Home

- Wi-Fi devices connect via a preset network identifier (SSID); hiding the SSID broadcast alone is **not** adequate security
- Change the default SSID and password; enable WPA2 encryption — but even WPA2 can still be vulnerable

KRACK (2017): a WPA2 protocol flaw exploited by Key Reinstallation Attacks — attackers break the encryption between a router and device to access network data. Affects all modern protected Wi-Fi; especially severe on Android 6.0+/Linux (tricked into installing an all-zero encryption key). Discovered by Mathy Vanhoef.

- Update all wireless-capable devices (routers, laptops, mobile) as soon as security patches are available
- Use a wired connection where possible
- Use a trusted VPN service on wireless networks

Bluetooth risk: can be exploited to eavesdrop, gain remote access, distribute malware, and drain batteries — keep Bluetooth off when not in use

3.1.4 Public Wi-Fi Risks

- Best not to access or send personal info on public Wi-Fi
- Verify the device isn't configured for file/media sharing and requires authenticated, encrypted access
- Use an encrypted VPN service to prevent eavesdropping — even if hackers intercept traffic in a VPN tunnel, they can't decipher it

3.1.5 Password Security

- Scenario: a colleague suggests reusing a personal password and keeping passwords written in a diary

Rating: 1/5 — Bad practice

3.1.6 A Strong Password

1. No dictionary words or names in any language
2. No common misspellings of dictionary words
3. Use special characters where possible (! @ # \$ % ^ & * ())
4. No computer or account names
5. Use a password with more than ten characters

3.1.7 Using a Passphrase

- A passphrase (e.g., "Acat th@tlov3sd0gs.") is longer and easier to remember than a password, and less vulnerable to dictionary/brute-force attacks

1. Choose a statement that is meaningful to you
2. Add special characters (! @ # \$ % ^ & * ())
3. The longer, the better

4. Avoid common or famous statements, e.g., song lyrics

3.1.8 Password Guidelines (NIST)

- US NIST published improved password requirements — intended for government, but a good standard generally:
- Passwords should be 8-64 characters
- Common, easily guessed passwords ('password', 'abc123') should not be used
- No forced composition rules (e.g., mandatory upper/lowercase + numbers)
- Users should be able to see the password while typing, to improve accuracy
- All printing characters and spaces should be allowed
- No password hints
- No forced password expiration period
- No knowledge-based authentication (secret questions, transaction history verification)

3.1.9 Password Check

Example: "**!Lik 3MySch001**" — rated a strong password

1 3.2 Data Maintenance

3.2.1 What Is Encryption?

- Converts information into a form unauthorized parties can't read; only someone with the secret key/password can decrypt it
- Encryption doesn't prevent interception — only prevents unauthorized viewing/access of the content
- Some criminals encrypt a victim's data and demand ransom to unlock it

3.2.2 How Do You Encrypt Your Data?

- Encrypting File System (EFS) is a Windows feature linked to a specific user account — only that user can access data encrypted with it

1. Select one or more files or folders
2. Right-click the selected data and go to 'Properties'
3. Find and click 'Advanced'
4. Select the 'Encrypt contents to secure data' check box
5. Encrypted files/folders are displayed in green in File Explorer

3.2.3 Back Up Your Data

- A backup may prevent loss of irreplaceable data
- Requires an additional storage location, with data copied there regularly

3.2.4 Are They Really Gone?

- Scenario: deleting old photos via drag-to-recycle-bin, then deleting from the bin

Correct answer: No — the photos are just inaccessible from the operating system, not actually gone

3.2.5 How Do You Delete Your Data Permanently?

- To erase data so it's unrecoverable, it must be overwritten with 1s and 0s multiple times, using dedicated tools — e.g., **SDelete** (Microsoft), **Shred** (Linux), **Secure Empty Trash** (Mac OS X)
- The only way to be *certain* data is unrecoverable is to physically destroy the hard drive/storage device — many criminals have exploited files thought to be "unrecoverable"
- Don't forget data stored online in the cloud — those copies also need to be deleted

2 3.3 Who Owns Your Data?

3.3.1 Terms of Service

- Scenario: accepting a Terms of Service without reading it, when setting up an online photo storage account

Correct definition: A Terms of Service is a legally binding contract that governs the rules of the relationship between you, the service provider, and others who use the service

3.3.2 Understand the Terms

- A Terms of Service includes sections from user rights and responsibilities to disclaimers and account modification terms

3.3.4 The Data Use Policy

- Sample clause: "you grant us a non-exclusive, transferable, sub-licensable, royalty-free, and worldwide license to host, use, distribute, modify, run, copy, publicly perform or display, translate, and create derivative works of your content"

Correct meaning: You own your content, but the photo sharing company can re-use any of your content for any purposes

3.3.5 Privacy Settings

- Scenario: default privacy settings applied because none were set before accepting the terms

Correct answer: Anyone will be able to see information about you and access your profile until you change the privacy settings

3.3.6 Before You Sign Up — Checklist

- Have you read the Terms of Service?
- What are your rights regarding your data?
- Can you request a copy of your data?
- What can the provider do with the data you upload?
- What happens to your data when you close your account?

3 3.4 Safeguarding Your Online Privacy

3.4.1 Two Factor Authentication

- Popular services (Google, Facebook, Twitter, LinkedIn, Apple, Microsoft) use 2FA to add an extra security layer for logins
- Besides username/password or PIN, 2FA requires a second token to verify identity:
- A physical object — credit card, mobile phone, or fob
- A biometric scan — fingerprint, facial, or voice recognition
- A verification code sent via SMS or email

Be careful! Even with 2FA, hackers can still gain access through phishing attacks, malware, and social engineering

3.4.2 Open Authorization (OAuth)

- OAuth is an open standard protocol that lets you use your credentials to access third-party applications without exposing your password

3.4.3 Social Sharing

- Scenario: updating a social profile with a new job position — deciding which missing fields to fill in

Safe to share: Your profile picture, Name of the organization. **Not safe:** email address, manager's/colleagues' contact info, date of birth, mobile phone number

3.4.4 Don't Get Spoofed

- A simple forged or spoofed email can lead to a massive data breach and irreversible reputational damage

3.4.5 Email and Web Browser Privacy

- Enabling private/in-private browsing mode minimizes tracking. Browser names for this mode:

Browser	Private Mode Name
Microsoft Internet Explorer	InPrivate
Google Chrome	Incognito
Mozilla Firefox	Private tab or private window
Safari	Private browsing

When private mode is enabled, cookies are disabled and temporary internet files/browsing history are cleared on close — helping prevent targeted-ad tracking. But even with private browsing and cookies disabled, companies constantly develop new fingerprinting methods to track users; intermediary devices like routers can also gather web surfing history.

4 3.5 Discover Your Own Risky Online Behavior

3.5.1 Scenario 1 — Social Media Sharing

What's safe to share on social media vs. what poses a risk — safe posting habits (e.g., "laughs with friends") vs. oversharing personal/location/financial details

3.5.2 Scenario 2 — Password Managers

- Reflective exercise: describe what a password manager is, how it works, and why it's beneficial

3.5.3 Scenario 3 — Open Wi-Fi

- A traveling sales manager forgot to download a contract; the only option is the train's open Wi-Fi

Correct advice: She should access the train's open Wi-Fi network and connect to the company servers using the VPN connection on her work laptop

3.5.4 Scenario 4 — Untrusted Downloads

- A manager suggests installing free software from an untrusted source to save money on a one-off project

Correct answer: No — the designer should check with a member of the IT team before installing any application from a non-trusted source

3.5.5 Scenario 5 — Scareware Pop-up

- A pop-up falsely claims an app needs an urgent update, pushing a "Download Now" button

1. Close the pop-up or browser by clicking 'X'
2. Protect the (uncorrupted) system by running a scan with trusted antivirus software
3. Check if a legitimate update is needed by visiting the official website directly

3.5.6 Scenario 6 — Phishing Emails

- Training exercise: spot the signs of an unsafe/phishing email claiming to be from inside the organization
- Red flags include urgent/pressuring language, vague or suspicious subject lines, and requests to "click this link quickly"

5 3.6 Quiz — Knowledge Check

Q1. What should you do to make sure people you live with do not have access to your secure data?

Ans — Set up password protection.

Q2. Which technology can prevent malicious software from monitoring activities, collecting personal info, and producing unwanted pop-up ads?

Ans — Antispyware.

Q3. Why do IoT devices pose a greater security risk than other computing devices on a network?

Ans — Most IoT devices do not receive frequent software updates.

Q4. Logging into a cloud storage account automatically gave access to a third-party printing service. What allowed this automatic authentication?

Ans — The cloud storage service is an approved application for the online printing service.

Q5. Which of the following is an example of two factor authentication?

Ans — Your fingerprint and your password.

Q6. How can you keep your personal browsing history hidden from other users on a shared computer?

Ans — Operate the web browser in private browser mode.

Q7. Which password would most likely take the longest for an attacker to guess or break?

Ans — mk\$\$citykat104#.

Q8. Which of the following is a requirement of a strong password?

Ans — Use special characters such as ! @ or \$.

Q9. Which wireless router configuration is NOT considered adequate security for a wireless network?

Ans — Preventing the broadcast of an SSID.

Q10. How can you prevent others from eavesdropping on network traffic on a public Wi-Fi hotspot?

Ans — Connect with a VPN service.

Q11. What is the only way of ensuring that deleted files on your computer are irrecoverable?

Ans — Physically destroying your computer's hard drive.

Q12. Which technology removes direct equipment and maintenance costs from the user for data backups?

Ans — A cloud service.

83%

Congratulations, you have passed the exam. ✓ Module 3 Knowledge Check — Complete

INTRODUCTION TO CYBERSECURITY

📌 MODULE 4 — PROTECTING THE ORGANIZATION

Cisco Networking Academy

Cisco Skills for All

Course Screenshots → Digital

COURSE

**Introduction to
Cybersecurity**

MODULE

4 of 5

STARTED

20 / 08 / 2026

FORMAT

Notes + Quiz + Test

0 4.1 Cybersecurity Devices and Technologies

4.1.1 Security Appliances

- No single security appliance solves all network security needs — orgs must combine multiple tools

Category	Role
Routers	Primarily interconnect network segments, but usually also provide basic traffic filtering
Firewalls	Look deeper into traffic to identify malicious behavior; apply sophisticated security policies to passing traffic
Intrusion Prevention Systems (IPS)	Use traffic signatures to match and block malicious traffic and attacks
Virtual Private Networks (VPN)	Let remote employees use a secure encrypted tunnel to connect back to the org's network; can also interconnect branch offices
Antimalware / antivirus	Use signatures or behavioral analysis to identify and block malicious code execution
Other security devices	Web and email security appliances, decryption devices, client access control servers, security management systems

4.1.2 Which Is It?

Appliance	Category
Cisco ISR 4000 — routing, filtering, encryption in a single platform	Router
Cisco Firepower 4100 Series — shows network activity to act faster against attacks	Firewall
Cisco AnyConnect Secure Mobility Client — secure remote access from any device/location	VPN
Cisco AMP — next-gen endpoint protection, scanning, constant file monitoring	Antimalware

Summary: ISR 4000 routers add traffic filtering, IPS, encryption and VPN capabilities. Firepower 4100 is a next-gen firewall with all ISR capabilities plus advanced network management/analytics. AnyConnect is Cisco's VPN client/server tech built into all Cisco security appliances. AMP is installed in next-gen routers, firewalls, IPS devices, web/email security appliances, or as host software.

4.1.3 Firewalls

Type	Description
Network layer firewall	Filters communications based on source and destination IP addresses
Transport layer firewall	Filters based on source/destination data ports and connection states
Application layer firewall	Filters based on an application, program, or service
Context aware layer firewall	Filters based on user, device, role, application type, and threat profile
Proxy server	Filters web content requests like URLs, domain names, and media types
Reverse proxy server	Placed in front of web servers; protects, hides, offloads, and distributes access to them
Network address translation (NAT) firewall	Hides or masquerades the private addresses of network hosts
Host-based firewall	Filters ports and system service calls on a single computer OS

4.1.4 Which One?

Scenario	Firewall Type
A small internal LAN needs internet access via a single connection	NAT firewall
Windows blocks access to apps on PCs from other network computers by default	Host-based firewall
Employees are blocked from specific URLs like gambling sites	Proxy server

4.1.5 Port Scanning

- Each app on a device is assigned a port number, used on both ends of a transmission to route data correctly
- Port scanning = probing a computer/server/host for open ports — can be malicious (reconnaissance) or harmless (verifying security policies)
- Tools: **Nmap / Zenmap** (official Nmap GUI) — free, multi-platform, scan profiles, command creator, searchable scan history

Response	Meaning
Open / Accepted	The port/service can be accessed by other network devices
Closed / Denied / Not Listening	The port/service isn't running — cannot be exploited
Filtered / Dropped / Blocked	Access is blocked by a firewall — cannot be exploited

To scan from outside a network, run the scan against the firewall/router's public IP (find it via "what is my IP address?"). Using an online tool like the Nmap Online Port Scanner, an open response on ports 21, 22, 25, 80, 443 or 3389 likely means port forwarding is enabled and servers are running on the private network.

4.1.6 What Does It Mean?

A port scan reports an 'open' state response — this means the service running on the network **can be accessed** by other network devices. Therefore, if the service contains a vulnerability, it **can be exploited** by an attacker.

4.1.7 Intrusion Detection and Prevention Systems

System	Description
IDS	Dedicated device or a tool on a server/firewall/host OS; scans data against a rules/signature database looking for malicious traffic. Logs detections and alerts an admin — does NOT take action (detect, log, report only). Usually placed offline (data mirrored via a switch) to avoid slowing the network (latency).
IPS	Placed inline (between switch and firewall); can block or deny traffic based on a positive rule/signature match — takes direct action.

One of the best-known IDS/IPS systems is **Snort**; its commercial version is **Cisco's Sourcefire**, which performs real-time traffic/port analysis, logging, content searching/matching, detects probes/attacks, executes port scans, and integrates with third-party reporting/analysis tools.

4.1.8 Real-Time Detection

- Hackers increasingly exploit software flaws before creators can fix them — a **zero-day attack**
- Goal: detect and stop such attacks in real time or within minutes
- Many organizations today can't detect attacks until days or months later
- Real-time detection requires active scanning via firewall/IDS/IPS devices, next-gen malware detection connected to global threat centers, and behavior/context-based anomaly detection

4.1.9 Protecting Against Malware

- DDoS is one of the biggest threats requiring real-time detection — hard to defend against since attacks come from hundreds/thousands of zombie hosts and look like legitimate traffic
- Defense against zero-day attacks and APTs: an enterprise malware detection solution like **Cisco's Advanced Malware Protection (AMP) Threat Grid** — client/server software analyzing millions of files against hundreds of millions of analyzed malware artifacts, giving a global view of malware attacks/campaigns

Team	Benefit from Threat Grid
Secure Operations Center team	Gathers more accurate, actionable data
Incidence Response team	Access to forensically sound info to quickly analyze suspicious behavior
Threat Intelligence team	Proactively improves the organization's security infrastructure
Security Infrastructure Engineering team	Consumes and acts on threat info faster, often automated

4.1.10 Security Best Practices

Guidelines drawn from repositories like the NIST Computer Security Resource Center:

- **Perform a risk assessment:** know the value of what you're protecting to justify security spend
- **Create a security policy:** clearly outline rules, job roles, responsibilities, and expectations for employees
- **Physical security measures:** restrict access to networking closets/server rooms; fire suppression
- **Human resources security measures:** background checks for all employees
- **Perform and test backups:** back up regularly and test data recovery
- **Maintain security patches and updates:** regularly update servers, clients, and network device OS/programs
- **Employ access controls:** configure user roles/privilege levels and strong authentication
- **Regularly test incident response:** employ an incident response team, test emergency scenarios
- **Implement a network monitoring, analytics and management tool:** integrates with other technologies
- **Implement network security devices:** next-gen routers, firewalls, and other appliances
- **Implement a comprehensive endpoint security solution:** enterprise-level antimalware/antivirus
- **Educate users:** provide security-procedure training — e.g., the SANS Institute for cybersecurity training/certifications
- **Encrypt data:** encrypt all sensitive organizational data, including email

4.2.1 Behavior-Based Security

- Captures and analyzes communication flow between a local-network user and a local/remote destination; changes in normal behavior patterns = anomalies, possibly an attack

Tool	Description
Honeypots	Lures an attacker in by appealing to their predicted malicious behavior; once inside, the admin captures, logs, and analyzes their behavior to build a better defense
Cisco's Cyber Threat Defense Solution Architecture	Uses behavior-based detection and indicators for greater visibility, context, and control — aims to know who's attacking, what type of attack, and where/when/how it's happening, using many security technologies together

4.2.2 NetFlow

- Gathers info about data flowing through a network — who/what devices are on it, and when/how users and devices access it
- Key component of behavior-based detection; NetFlow-enabled switches/routers/firewalls report data entering, leaving, and traveling through the network
- Data goes to NetFlow collectors, which establish baseline behaviors across 90+ attributes (e.g., source/destination IP, ports, protocol, bytes/packet)

4.2.3 Penetration Testing

- Pen testing = assessing a system/network/organization for security vulnerabilities by attempting to breach systems, people, processes, and code — findings improve future defenses

Step	Description
1. Planning	Gather as much info as possible on the target — passive/active reconnaissance (footprinting) and vulnerability research
2. Scanning	Active reconnaissance to identify weaknesses — port scanning, vulnerability scanning, enumeration (identifying user/system/admin accounts)
3. Gaining access	Attempt exploitation — launching exploits/payloads, breaching physical barriers, social engineering, exploiting website/software/hardware vulnerabilities or misconfigurations, breaching access controls, cracking weak Wi-Fi
4. Maintaining access	Stay undetected using backdoors, Trojans, rootkits, and covert channels while gathering valuable data
5. Analysis and reporting	Provide a report recommending updates to products, policies, and training to improve security

4.2.5 Impact Reduction — Actions After a Security Breach

- **Communicate the issue:** creates transparency — inform employees internally, clients externally via direct communication/official announcements
- **Be sincere and accountable:** respond honestly, take responsibility where the org is at fault
- **Provide the details:** explain why the breach happened and what was compromised; cover client costs for identity theft services if needed
- **Find the cause:** hire forensics experts to research what caused/facilitated the breach
- **Apply lessons learned:** apply forensic findings to prevent similar breaches
- **Check, and check again:** ensure all systems are clean — no backdoors or further compromise (attackers often leave backdoors for future breaches)
- **Educate!:** raise awareness and train employees, partners, and clients to prevent future breaches

4.2.6 What Is Risk Management?

- Formal, continuous process of identifying/assessing risk to reduce impact of threats/vulnerabilities — risk can't be eliminated, only managed to an acceptable level; a control's cost should never exceed the asset's value

Step	Description
Frame the risk	Identify threats that increase risk — processes, products, attacks, service disruption, reputational damage, legal liability, IP loss
Assess the risk	Determine severity of each threat — prioritize via quantitative (financial impact) or qualitative (operational impact) analysis
Respond to the risk	Develop an action plan — eliminate, mitigate, transfer, or accept each risk
Monitor the risk	Continuously review risk reduced through action; closely monitor any accepted (non-eliminated) risks

4.3.1 Cisco's CSIRT

- Computer Security Incident Response Team (CSIRT) receives, reviews, and responds to security incident reports
- Cisco's CSIRT goes further — proactive threat assessment, mitigation planning, incident trend analysis, and security architecture review to prevent incidents before they happen
- Collaborates with FIRST, NSIE, DSIE, and DNS-OARC to stay current on new developments
- National/public CSIRT organizations that help develop incident management capabilities: US-CERT, CERT-EU, FIRST, APCERT, and the Software Engineering Institute (Carnegie Mellon University)

4.3.2 Security Playbook

Organizations should provide guidance on:

- How to identify cybersecurity risk to systems, assets, data, and capabilities
- Implementation of safeguards and personnel training
- A flexible response plan minimizing impact/damage during a breach
- Security measures/processes needed in the aftermath of a breach

A **security playbook** is a collection of repeatable queries/reports outlining a standardized incident detection and response process. It should:

- Highlight how to identify/automate responses to common threats (malware-infected machines, suspicious network activity, irregular authentication attempts)
- Clearly define inbound and outbound traffic
- Provide summary info — trends, statistics, and counts
- Give quick access to key statistics and metrics
- Correlate events across all relevant data sources

4.3.3 Tools for Incident Detection and Prevention

Tool	Description
SIEM (Security Information and Event Management)	Collects and analyzes security alerts, logs, and other real-time/historical data from network security devices to enable early attack detection
DLP (Data Loss Prevention)	Stops sensitive data from being stolen or escaping a network; monitors/protects data in 3 states — in use, in motion, and at rest

4.3.4 Cisco's ISE and TrustSec

- Cisco Identity Services Engine (ISE) and TrustSec enforce user access to network resources by creating **role-based access control policies**

4.3.5 Talk the Talk — Key Terms Recap

- **IPS:** blocks/denies traffic based on a positive rule or signature match
- **IDS:** scans data against a rules/attack-signature database, looking for malicious traffic
- **DLP:** stops sensitive data from being stolen or escaping a network
- **SIEM:** collects and analyzes security alerts, logs, and real-time/historical data from network security devices

3 4.4 Quiz — Knowledge Check

Q1. Which tool can be used to provide a list of open ports on network devices?

Ans — Nmap.

Q2. "Today, there are single security appliances that will solve all the network security needs of an organization." True or false?

Ans — False.

Q3. What tool can identify malicious traffic by comparing packet contents to known attack signatures?

Ans — IDS.

Q4. Which tool can perform real-time traffic and port analysis, and also detect port scans, fingerprinting, and buffer overflow attacks?

Ans — Snort.

Q5. What is the last stage of a pen test?

Ans — Analysis and reporting.

Q6. Behavior-based analysis involves using baseline information to detect what?

Ans — Anomalies.

Q7. Which actions should an organization take in the event of a security breach? (select two)

Ans — Communicate a call to action to all employees; Carry out research to uncover what caused the breach.

Q8. How do Cisco ISE and TrustSec work?

Ans — They enforce access to network resources by creating role-based access control policies.

INTRODUCTION TO CYBERSECURITY

♥ MODULE 5 — WILL YOUR FUTURE BE IN CYBERSECURITY?

Cisco Networking Academy

Cisco Skills for All

Course Screenshots → Digital

COURSE	MODULE	STARTED	FORMAT
Introduction to Cybersecurity	5 of 5	21 / 08 / 2026	Notes + Quiz

0 5.1 Legal and Ethical Issues

5.1.1 Legal Issues in Cybersecurity

- Cybersecurity professionals need the **same skills as attackers**, but must use them strictly **within the bounds of the law**

Personal legal issues:

- Having the opportunity/skill to hack someone's computer or network doesn't mean you should — "just because you can does not mean you should"
- Most hacks leave tracks, which can be traced back to you
- Skills can be used positively or illegally — huge demand exists for professionals who put cyber skills to good, legal use

Corporate legal issues:

- Most countries have cybersecurity laws that businesses/organizations must follow
- Breaking these laws at work can get the organization punished and you could lose your job — in other cases you could be prosecuted, fined, or sentenced
- Rule of thumb:** if unsure whether an action/behavior might be illegal, assume it is illegal and don't do it — always check with the legal or HR department

International law and cybersecurity:

- International cybersecurity law is a constantly evolving field
- Cyberattacks happen in cyberspace, which has no traditional geographic boundaries
- It's much easier to mask the source of an attack in cyberwarfare than in conventional warfare
- International law is shaped by country practice, *opinio juris* (a country's sense that it is bound to the law in question), and any treaties drafted

5.1.2 Ethical Issues in Cybersecurity

- Besides staying within the law, you must also demonstrate **ethical behavior** and act responsibly in your work

Scenario: A pen test carried out for @Apollo revealed that a colleague, who started at the same time as you, was responsible for a data breach. You are thinking of leaving this out of your report so they don't get in trouble.

Questions to help decide the best course of action:

- Is it legal?
- Does your action comply with @Apollo policy?
- Will your action be favorable for @Apollo and its stakeholders?
- Would it be okay if everyone at @Apollo took this action?
- Would the outcome of your action represent @Apollo in a positive light in a news headline?

5.1.3 How Did You Answer?

If "Yes" to all questions: likely appropriate to move forward with the action — but remember, just because something is legal doesn't mean it's ethical. In this case, withholding the breach info from the pen test report isn't illegal, but it isn't the ethical thing to do — not reporting it could be devastating for @Apollo and its customers.

If "No" to any question: stop and reconsider — it could have serious legal consequences for you and the organization. Every new finding in cybersecurity **must** be reported to protect the organization and its customers. Always seek advice from your line manager, legal, or HR if unsure whether an action/behavior might be unethical.

- An **Ethics Decision Tree** is available (as a downloadable resource) that can be applied to your own place of work

5.1.4 Corporate Ethical Issues

- Many professional IT organizations, such as the **Information Systems Security Association (ISSA)**, publish Codes of Ethics to guide employee actions and behaviors
- Cisco has a team devoted exclusively to ethical business conduct, plus a Code of Business Conduct to help employees make business decisions and resolve issues

1 5.2 Education and Careers

5.2.1 Become a Cybersecurity Guru

- After excelling as an entry-level Security Analyst, there are many cybersecurity jobs to consider
- Even early in a cybersecurity career, it's a good idea to browse job search engines — Indeed, LinkedIn, Monster, and CareerBuilder — to see what kinds of jobs are available worldwide

5.2.2 Professional Certifications

Certifications verify skills/knowledge and can boost your career:

Certification	What it covers
Cisco Certified Support Technician (CCST) Cybersecurity	Entry-level cert for newcomers starting a cybersecurity career; aimed at high school/early college students and career changers; does not expire or require periodic recertification
CompTIA Security+	Entry-level security cert that meets U.S. Dept. of Defense Directive 8570.01-M requirements — important for working in IT security for the federal government
EC-Council Certified Ethical Hacker (CEH)	Tests understanding of finding weaknesses/vulnerabilities in target systems using the same knowledge/tools as a malicious hacker, but in a lawful, legitimate manner
ISC2 Certified Information Systems Security Professional (CISSP)	Most recognizable and popular security certification; requires at least 5 years of relevant industry experience to take the exam
Cisco Certified CyberOps Associate	Validates skills required of associate-level cybersecurity analysts within Security Operations Centers (SOC) — directly relevant to SOC L1 roles

5.2.3 Cybersecurity Career Pathways

- Depending on personal interests, different job options exist to explore — some require specialized skills and professional certifications
- CyberSeek** is a tool providing detailed supply/demand data for the cybersecurity job market, to help close the cybersecurity skills gap
- Its interactive career pathway shows the range of cybersecurity jobs plus salary, credential, and skillset info for each

Level	Roles
Feeder Roles	Networking, Software Development, Systems Engineering, Financial and Risk Analysis, Security Intelligence
Entry-Level	Cybersecurity Specialist/Technician, Cyber Crime Analyst/Investigator, Incident Analyst/Responder, IT Auditor
Mid-Level	Cybersecurity Analyst, Cybersecurity Consultant, Penetration & Vulnerability Tester
Advanced-Level	Cybersecurity Manager/Administrator, Cybersecurity Engineer, Cybersecurity Architect

Feeder roles are the backgrounds professionals often come from before moving into cybersecurity; the pathway then progresses Entry → Mid → Advanced level.

2 5.3 Quiz — Knowledge Check

Q1. What is the difference between a hacker and a cybersecurity professional?
Ans — Cybersecurity professionals must work within legal boundaries.

Q2. What can the skills developed by cybersecurity professionals be used for?
Ans — Cybersecurity professionals develop many skills that can be used for good or evil.

100%

Congratulations, you have passed the exam. ✓ Module 5 Knowledge Check — Complete

INTRODUCTION TO CYBERSECURITY

🔑 COURSE FINAL EXAM — ANSWER KEY

Cisco Networking Academy

15 Questions

Verified Correct Answers Only

COURSE

Introduction to Cybersecurity

EXAM TYPE

Course Final Exam

ATTEMPTED

20 / 09 / 2026

FORMAT

15 Q — Verified Key

Note: A couple of answers picked during the live attempt were incorrect — this answer key has been corrected against verified course content. Only the correct answer is listed for each question.

✓ Final Exam — Questions & Verified Answers

Q1. You are configuring access settings to require employees to authenticate first before accessing certain web pages. Which requirement of information security is addressed?

Ans — Confidentiality.

Q2. Which of the following items are states of data? (Choose three)

Ans — Storage, Processing, Transmission.

Q3. What is the main purpose of cyberwarfare?

Ans — To gain advantage over adversaries.

Q4. An organization's IT department reports an abnormally high number of web page requests from different locations simultaneously. What type of security attack is occurring?

Ans — DDoS.

Q5. What do you call the vulnerabilities discovered by Google security researchers that affect almost all CPUs released since 1995? (Select two)

Ans — Meltdown, Spectre.

Q6. "Cryptocurrencies are handled on a centralized exchange." Is this statement true or false?

Ans — False.

Q7. Which technology creates a security token that allows a user to log in to a desired web application using credentials from a social media website?

Ans — Open authorization (OAuth).

Q8. What is the best way to avoid getting spyware on a machine?

Ans — Install software only from trusted websites.

Q9. A medical office employee sends emails to patients about their recent visits. What information would put the privacy of the patients at risk if included in the email?

Ans — Patient records.

Q10. "After a data breach, it's important to educate employees, partners and customers on how to prevent future breaches." True or false?

Ans — True.

Q11. Which of the following firewalls filters ports and system service calls on a single computer operating system?

Ans — Host-based firewall.

Q12. What is the main function of the Cisco Security Incident Response Team (CSIRT)?

Ans — To ensure company, system and data preservation.

Q13. One of your colleagues has lost her identification badge. She is in a hurry for a meeting and doesn't have time to visit HR for a temporary badge. You lend her your identification badge until she can obtain a replacement. Is this behavior ethical or unethical?

Ans — Unethical.

Q14. Which certification tests your understanding of finding weaknesses/vulnerabilities in target systems using the same knowledge and tools as a malicious hacker, but in a lawful and legitimate manner?

Ans — EC-Council Certified Ethical Hacker (CEH).

Q15. Which certification does not expire or require periodic recertification, and is geared towards post-secondary graduates and those interested in a career change?

Ans — ISACA CSX Cybersecurity Fundamentals.

INTRODUCTION TO CYBERSECURITY

🎓 COURSE COMPLETED — CERTIFICATE & BADGE

Cisco Networking Academy

Verified Achievement

COURSE	COMPLETION DATE	TOTAL HOURS	ACADEMY
Introduction to Cybersecurity	20 / 09 / 2026	6	Cisco Networking Academy



Certificate of Completion

Cisco
Networking
Academy

This certificate is awarded to

Shivam Kumar

for successfully completing

Introduction to Cybersecurity

offered by Networking Academy
through the Cisco Networking Academy program.

Lynn Bloomer

Lynn Bloomer
Director
Cisco Networking Academy



20 Sep 2026
Completion Date

Cert ID: 800f2289-ffe5-449b-af83-cbc2cde8ed75

Awarded to Shivam Kumar for successfully completing "Introduction to Cybersecurity", offered by Networking Academy through the Cisco Networking Academy program — Cert ID: 800f2289-ffe5-449b-af83-cbc2cde8ed75



Verified Skill Badge

Introduction to Cybersecurity



Introduction to Cybersecurity

Explore the exciting field of cybersecurity and why cybersecurity is a future-proof career.

ISSUED DATE

Sep 20, 2026

TOTAL HOURS

6

ACADEMY

Networking Academy

SKILLS YOU LEARN

Introduction to Cybersecurity

[View & Share Badge](#)

Introduction to Cybersecurity — Verified Badge | Issued Sep 20, 2026 | Total Hours: 6 | Academy: Networking Academy | Skill: Introduction to Cybersecurity

✓ **Complete**

Introduction to Cybersecurity — Course fully completed, certified, and badged. Onward to the next milestone on the SOC L1 roadmap!

Ethics and Security

ISSA International Ethics Committee

Importance of Ethics to Security

- Information Security professionals are entrusted with the crown jewels of an organization.
- Ethical behavior, both on and off-the-job, is the assurance that we are worthy of that trust.
- IS Security sets and upholds a standard
 - Corporate Ethics programs originating from the CSO
 - Promote uniform adherence to policy through example

Topics

- Ethics in the Information Security Realm
- ISSA International Ethics Posture
- ISSA International Ethics Committee
- Importance of Ethics To Security
- Responsibilities of Security Professionals

Ethics Overview

- Ethics is about how we ought to live*
- The purpose of Ethics in Information Security is not just philosophically important, it can mean the survival of a business or an industry**

*Ethics is doing the right thing,
even when no one is looking*

ISSA International Code of Ethics (Part 1)

- Perform all professional activities and duties in accordance with all applicable laws and the highest ethical principles;
- Promote generally accepted information security current best practices and standards;
- Maintain appropriate confidentiality of proprietary or otherwise sensitive information encountered in the course of professional activities;

ISSA International Code of Ethics (Part 2)

- Discharge professional responsibilities with diligence and honesty;
- Refrain from any activities which might constitute a conflict of interest or otherwise damage the reputation of employers, the information security profession, or the Association; and
- Not intentionally injure or impugn the professional reputation or practice of colleagues, clients, or employers.

ISSA's Posture – Ethics for the Security Professional

- Set Ethical Standards for Membership
 - Include Broader Audiences
- Educated and Informed Members
 - Case Studies, Articles, Courses
- Universally Applicable Standards
 - Geographically, Culturally
 - Cross Discipline

ISSA International Ethics Committee

- Founded in 2002
- 15 active members
- Purpose: Provide guidance on ethical behavior for Information System Security professionals, develop and maintain guidelines for ethics relating to Information Security practices.

***Proactive Promotion and Education
to Influence Positive Behavior***

Accomplishments

- Approved policy by ISSA International Board
 - Reporting and reviewing ethical complaints, appeals
- Respond to and hear valid ethics complaints
 - Time-sensitive
 - Confidential
 - Unbiased
 - Consistent analysis of facts and perspectives
 - Findings referred up to ISSA International Board
- New Disclosure of Relationships Process
 - Identify and mitigate potential Conflicts of Interest
 - Completed forms are reviewed and suggestions provided
 - ISSA International Board, ISSA Foundation, Ethics Committee
- Articles for ISSA Journal, Outreach and Education
- Ad-hoc research

ISSA Ethics Complaint Handling

- Formal, Written Complaint is Received and Verified for Completeness
- Notices sent to both parties
 - Complete Complaint
 - Copy of Policy, Clear Description of Next Steps
 - Listing of Ethics Committee members (ability to recuse members – eliminate bias)
- Evaluation of Facts as Submitted by Both Parties
 - Some Clarification may be Requested
 - Mediation Assistance may be Requested
- Hearing Panel Assembled – Conference Call Scheduled
 - At least 3 members of the Committee (Voting)
 - A member of the ISSA International Board (Voting)
 - Include a current Chapter Officer (Voting)
 - Association Attorney (Non-Voting)
- Findings and Recommendation Sent to ISSA International Board

Ethical Challenges in InfoSec

- Misrepresentation of certifications, skills
- Abuse of privileges
- Inappropriate monitoring
- Withholding information
- Divulging information inappropriately
- Overstating issues
- Conflicts of interest
- Management / employee / client issues

Ethical Challenges – Snake Oil

- “Consultants” who profess to offer information security consulting, but offer profoundly bad advice
- “Educators”, both individuals and companies, that offer to teach information security, but provide misinformation (generally through ignorance, not intent)
- “Security Vendors”, who oversell the security of their products
- “Analysts”, who oversimplify security challenges, and try to upsell additional services to naïve clients
- “Legislators”, who push through “from-the-hip” regulations, without thoughtful consideration of their long-term impact

Some Resource Links

<http://ethics.csc.ncsu.edu/>

<http://www.ethicsweb.ca/resources/>

<http://ethics.iit.edu/index.html>

<http://onlineethics.org/>

On the development of a personal code of ethics...

http://www.domain-b.com/management/general/20060401_personal.html

Corporate ethics training...

<http://www.integrity-interactive.com/>

<http://www.easyi.com/enus/business-ethics/solutions.asp>

On the role of ethics...

http://securityawareness.blogspot.com/2005/07/role-of-ethics-in-information-security_07.html

http://www.seifried.org/security/index.php/Closet20000531_Ethics_in_Information_Security

Something from the SANS Reading Room

<http://www.sans.org/rr/whitepapers/legal/54.php>

Ten Commandments of Ethics in Information Security

Thou shalt not use a computer to harm other people.

Thou shalt not interfere with other people's computer work.

Thou shalt not snoop around in other people's computer files.

Thou shalt not use a computer to steal.

Thou shalt not use a computer to bear false witness.

Thou shalt not copy or use proprietary software for which you have not paid.

Thou shalt not use other people's computer resources without authorization or proper compensation.

Thou shalt not appropriate other people's intellectual output.

Thou shalt think about the social consequences of the program you are writing or the system you are designing.

Thou shalt always use a computer in ways that insure consideration and respect for your fellow humans.

Questions/Discussion